Semiring-Based Linear Algebra Framework for Two-Way Regular Path Queries

Georgiy Belyanin[1[0009−0004−7527−4281]], Rodion Suvorov[1[!!!!]], and Semyon Grigoriev[1[0000−0002−7966−0698]]

Saint Petersburg State University, Saint Petersburg, Russia belyaningeorge@ya.ru, !!!!, s.v.grigoriev@mail.spbu.ru

Abstract. Two-way regular path queries (2-RPQs) allow the use of regular languages over edges and their inverses in edge-labeled graphs to constrain paths of interest. Adopted in modern graph query languages such as GQL, SQL/PGQ, and Cypher, 2-RPQs have become part of the GQL ISO standard. However, their efficient evaluation on large-scale graphs remains challenging. In this paper, we propose a semiring-based generalization of the linear algebra approach for evaluating single-source 2-RPQs. The original LARPQ algorithm solves the reachability problem using Boolean matrices. We generalize it by replacing Boolean algebra with arbitrary semirings, enabling a unified solution for various path-aggregation problems. Specifically, we employ the path semiring to find all simple paths and all trails. The algorithms are implemented using the SuiteSparse:GraphBLAS library within the LAGraph infrastructure. Experimental evaluation on real-world knowledge graphs demonstrates significant performance improvements over state-of-the-art solutions.

Keywords: Regular path queries · Two-way RPQ · Linear algebra · Semirings · Graph databases · Sparse matrices.

1 Introduction

Language-constrained path querying (FLPQ) [1] provides a mechanism for searching paths in edge-labeled graphs where constraints are specified in terms of formal languages. A path is considered valid if the word formed by concatenating the labels of its edges belongs to the specified language. When constraints are regular languages, such queries are called regular path queries (RPQs). In practice, it is often useful to allow traversal opposite to edge directions, leading to two-way regular path queries (2-RPQs) [**?**,**?**]. Queries specifying only a start or an end vertex are known as single-source or single-destination queries, respectively. Both have been adopted in graph query languages such as GQL, SQL/PGQ [**?**], and SPARQL [**?**].

Graph database systems support different path query semantics, as the choice significantly affects both result interpretation and computational complexity. Common semantics include: reachability (all vertices reachable via any valid path), all-shortest-paths (one shortest path per reachable vertex), all-simple-paths (all simple paths satisfying the query), and all-trails (all trails satisfying

the query) [**?**]. Each semantics serves different use cases, and the query engine must select the appropriate algorithm accordingly.

Efficient evaluation of RPQs and 2-RPQs remains an active research area [**?**,**?**]. Sparse linear algebra based methods have shown promise for RPQ evaluation [**?**]. In particular, the LARPQ (Linear Algebra-based Regular Path Query) algorithm [**?**] employs a breadth-first traversal that simultaneously processes both the graph and the finite automaton representing the regular language. While LARPQ demonstrates promising performance, it was designed solely for reachability, and its application to queries with fixed destination vertices remains unexplored.

The contributions of this paper are as follows.

- A formal generalization of the LARPQ algorithm to a semiring-based framework, enabling the solution of various path-aggregation problems within a unified algorithmic structure.
- Instantiation of the framework with specific semirings to solve distinct problems: (i) Boolean semiring for reachability, and (ii) path semiring with index unary operators for all-simple-paths and all-trails semantics.
- ??? Proof of correctness and computational complexity analysis for the generalized algorithm.
- Experimental evaluation on real-world knowledge graphs demonstrating significant performance improvements over state-of-the-art alternatives.

2 Preliminaries

In this section, we introduce the foundational concepts required throughout the paper, including edge-labeled graphs, paths, finite automata, and regular path queries.

2.1 Edge-Labeled Graphs

We begin by defining the graph data model used in this work.

Definition 1 (Edge-labeled graph). *A quadruple $G = \langle V, E, L, \lambda_G \rangle$ is called an edge-labeled graph (or simply a graph) if:*

- *V is a finite set of vertices;*
- *$E \subseteq V \times V$ is a finite set of edges;*
- *L is a finite set of labels;*
- *$\lambda_G : E \mapsto 2^L$ represents edge labels.*

For the remainder of this paper, we assume that vertices of any graph $G = \langle V, E, L, \lambda_G \rangle$ are enumerated, and without loss of generality, $V = \{1, 2, \ldots, |V|\}$.

To enable traversal in both directions along edges, we introduce the notion of inverted edges and labels. Let $G = \langle V, E, L, \lambda_G \rangle$ be an edge-labeled graph. We define:

- $a^- \notin L$ for each $a \in L$, with $(a^-)^- = a$ and $a = b \Leftrightarrow a^- = b^-$;
- $e^- = (v, u)$ where $e = (u, v)$;
- $E^- = \{e^- \mid e \in E\}$;
- $L^- = \{a^- \mid a \in L\}$;
- $\lambda_G^- : E^- \mapsto 2^{L^-}$, where $\lambda_G^-(e^-) = \{a^- \mid a \in \lambda_G(e)\}$.

We also define the bidirectional edge and label sets:

- $E^{\leftrightarrow} = E \cup E^-$;
- $L^{\leftrightarrow} = L \cup L^-$;
- $\lambda_G^{\leftrightarrow} : E^{\leftrightarrow} \mapsto 2^{L^{\leftrightarrow}}$, where $\lambda_G^{\leftrightarrow}(e) = \lambda_G(e) \cup \lambda_G^-(e)$;
- $G^{\leftrightarrow} = \langle V, E^{\leftrightarrow}, L^{\leftrightarrow}, \lambda_G^{\leftrightarrow} \rangle$.

The graph $G^{\leftrightarrow}$ allows traversal along both original and inverted edges, which is essential for evaluating two-way regular path queries.

2.2 Paths

Definition 2 (Path). *A path $\pi = (e_1, e_2, \ldots, e_n)$ of length n in the graph $G = \langle V, E, L, \lambda_G \rangle$ from u_1 to v_n is a finite sequence of edges $e_i = (u_i, v_i) \in E$ such that $\forall 1 \leq i \leq n-1 : v_i = u_{i+1}$.*

We also consider zero-length paths, represented by an empty sequence from a vertex to itself.

Definition 3 (Simple path). *A path $\pi = (e_1, e_2, \ldots, e_n)$ is simple if it contains no repeated vertices except possibly the first and last. Formally, $\forall 1 \leq i, j \leq n : v_i = u_j \Rightarrow j = i + 1$.*

Definition 4 (Trail). *A path $\pi = (e_1, e_2, \ldots, e_n)$ is a trail if it contains no repeated edges. Formally, $\forall 1 \leq i, j \leq n : e_i = e_j \Rightarrow i = j$.*

Definition 5 (Two-way path). *A two-way path (2-path) $\pi = (e_1, e_2, \ldots, e_n)$ in G is a path in $G^{\leftrightarrow}$. Two-way simple paths and two-way trails are defined analogously.*

The mapping ω associates paths with words over the label alphabet:

- $\omega_G(\pi) = \{a_1 \cdot a_2 \cdot \ldots \cdot a_n \mid a_i \in \lambda_G(e_i)\}$ for a path $\pi = (e_1, e_2, \ldots, e_n)$ in G;
- $\omega_{G^{\leftrightarrow}}(\pi) = \{a_1 \cdot a_2 \cdot \ldots \cdot a_n \mid a_i \in \lambda_G^{\leftrightarrow}(e_i)\}$ for a 2-path $\pi = (e_1, e_2, \ldots, e_n)$ in G.

Here, $\cdot$ denotes string concatenation.

2.3 Two-Way Finite Automata

Regular languages, recognized by finite automata, provide the foundation for specifying path constraints. For two-way path queries, we require a variant of nondeterministic finite automata.

Definition 6 (Two-way NFA). *A two-way nondeterministic finite automaton (2-NFA) is a tuple $N = \langle Q, \Sigma, \Delta, \lambda_N, Q_S, Q_F \rangle$, where:*

- *Q is a finite set of states;*
- *Σ is a finite alphabet;*
- *$\Delta \subseteq Q \times Q$ is the transition relation;*
- *$\lambda_N : \Delta \mapsto 2^{\Sigma^{\leftrightarrow}}$ maps transitions to labels;*
- *$Q_S \subseteq Q$ is the set of starting states;*
- *$Q_F \subseteq Q$ is the set of final states.*

Definition 7 (Two-way DFA). *A two-way deterministic finite automaton (2-DFA) is a special case of 2-NFA with $|Q_S| = 1$ and $\forall (q, q'), (q, q'') \in \Delta : q' \neq q'' \Rightarrow \lambda_N((q, q')) \cap \lambda_N((q, q'')) = \emptyset$.*

A 2-NFA can be viewed as a graph $G_N = \langle Q, \Delta, \Sigma^{\leftrightarrow}, \lambda_N \rangle$ equipped with sets of starting and final states. Conversely, a graph $G = \langle V, E, L, \lambda_G \rangle$ equipped with $V_S \subseteq V$ can be seen as a 2-NFA $N_G = \langle V, L, E, \lambda_G^{\leftrightarrow}, V_S, V \rangle$ that accepts:

$$[[N_G]] = \bigcup_{\text{2-path } \pi_G \text{ in } G \text{ from } v_S \in V_S} \omega_{G^{\leftrightarrow}}(\pi_G)$$

2.4 Two-Way Regular Path Queries

Definition 8 (Two-way regular path query). *A two-way regular path query (2-RPQ) is a triple $\langle G, R, v_s \rangle$ where $G = \langle V, E, L, \lambda_G \rangle$ is a graph, R is a regular language over the alphabet $L^{\leftrightarrow}$, and $v_s \in V$ is a starting vertex.*

We define several semantic variants of 2-RPQs that are considered in this paper:

- **Single-source reachability (SSR):**

  $$[[Q]]_{SSR} = \{u \in V \mid \exists \text{ 2-path } \pi \text{ in } G \text{ from } v_s \text{ to } u \text{ and } \omega_{G^{\leftrightarrow}}(\pi) \cap R \neq \emptyset\}.$$

- **Single-destination reachability (SDR):**

  $$[[Q]]_{SDR} = \{u \in V \mid \exists \text{ 2-path } \pi \text{ in } G \text{ from } u \text{ to } v_s \text{ and } \omega_{G^{\leftrightarrow}}(\pi) \cap R \neq \emptyset\}.$$

- **Single-source all simple paths (SSASP):**

  $$[[Q]]_{SSASP} = \{\pi \mid \pi \text{ is a 2-simple path in } G \text{ from } v_s \text{ and } \omega_{G^{\leftrightarrow}}(\pi) \cap R \neq \emptyset\}.$$

- **Single-destination all simple paths (SDASP):**

  $$[[Q]]_{SDASP} = \{\pi \mid \pi \text{ is a 2-simple path in } G \text{ to } v_s \text{ and } \omega_{G^{\leftrightarrow}}(\pi) \cap R \neq \emptyset\}.$$

– **Single-source all trails (SSAT):**

$$[[Q]]_{SSAT} = \{\pi \mid \pi \text{ is a 2-trail in } G \text{ from } v_s \text{ and } \omega_{G^{\leftrightarrow}}(\pi) \cap R \neq \emptyset\}.$$

– **Single-destination all trails (SDAT):**

$$[[Q]]_{SDAT} = \{\pi \mid \pi \text{ is a 2-trail in } G \text{ to } v_s \text{ and } \omega_{G^{\leftrightarrow}}(\pi) \cap R \neq \emptyset\}.$$

3 Libear Algebra, Graphs And Relations

In this section, we establish the linear algebra foundations required for our approach. We introduce the concept of semirings, discuss matrix representations of relations, and present the original LARPQ algorithm for solving the reachability problem.

3.1 Semirings

Semirings provide an algebraic framework that generalizes both Boolean algebra and path-based computations. A semiring consists of a set equipped with two binary operations that satisfy certain axioms.

Definition 9 (Semiring). *A semiring is a tuple $\langle S, \oplus, \otimes, \mathbf{0}, \mathbf{1}\rangle$ where:*

- $\langle S, \oplus, \mathbf{0}\rangle$ *is a commutative monoid with identity* $\mathbf{0}$*;*
- $\langle S, \otimes, \mathbf{1}\rangle$ *is a monoid with identity* $\mathbf{1}$*;*
- $\otimes$ *distributes over* $\oplus$*:* $a \otimes (b \oplus c) = (a \otimes b) \oplus (a \otimes c)$ *and* $(a \oplus b) \otimes c = (a \otimes c) \oplus (b \otimes c)$*;*
- $\mathbf{0}$ *is an annihilator for* $\otimes$*:* $a \otimes \mathbf{0} = \mathbf{0} \otimes a = \mathbf{0}$.

Two specific semirings are particularly important for this work: the Boolean semiring for reachability and the path semiring for enumerating paths.

3.2 Matrix Representations

Relations between finite sets can be represented using matrices. For two enumerated finite sets $A = \{1, 2, \ldots, n\}$ and $B = \{1, 2, \ldots, m\}$, a binary matrix T of size $n \times m$ can represent a relation $T \subseteq A \times B$ where $T_{ij} = 1 \Leftrightarrow (i, j) \in T$.

Let $G = \langle V, E, L, \lambda_G\rangle$ be an arbitrary graph and $N = \langle Q, \Sigma, \Delta, \lambda_N, Q_S, Q_F\rangle$ an arbitrary 2-NFA. For each label $a \in L^{\leftrightarrow}$, we define:

- $E_a = \{e \mid e \in E, a \in \lambda_G^{\leftrightarrow}(e)\}$;
- $\Delta_a = \{\delta \mid \delta \in \Delta, a \in \lambda_N(\delta)\}$.

Definition 10 (Adjacency matrix of a label). *Let G_a be the matrix representing the binary relation E_a. Then G_a is called the adjacency matrix of label a.*

Definition 11 (Boolean decomposition). *A Boolean decomposition of the adjacency matrix G is the set of Boolean matrices $\{G_a \mid a \in L^{\leftrightarrow}\}$.*

Note that for a given graph $G = \langle V, E, L, \lambda_G\rangle$, we have $E_{a^-} = (E_a)^-$ and $G_{a^-} = (G_a)^T$, enabling efficient traversal in both directions.

3.3 Boolean Matrix Operations

For matrices over the Boolean semiring $\langle \mathbb{B} = \{0,1\}, \vee, \wedge, \neg, 0, 1\rangle$, we define the following operations. Let $A = (a_{ij})$, $B = (b_{ij})$, $C = (c_{ij})$ be matrices:

- **Sum:** $A_{n\times m} + B_{n\times m} = (a_{ij} \vee b_{ij})_{n\times m}$;
- **Product:** $A_{n\times m} \times B_{m\times k} = (\bigvee_{l=1}^{m} a_{il} \wedge b_{lj})_{n\times k}$;
- **Transposition:** $A^T_{n\times m} = (a_{ji})_{m\times n}$;
- **Zero-matrix:** $0_{n\times k} = (0)_{n\times k}$;
- **Mask:** $A_{n\times m}\langle C_{n\times m}\rangle = (a_{ij} \wedge c_{ij})_{n\times m}$;
- **Complement:** $\neg A = (\neg a_{ij})$.

These operations suffice for expressing the reachability algorithm.

3.4 The Path Semiring

For solving path enumeration problems (all simple paths, all trails), we require a more expressive semiring based on path representations.

Let $RPaths = 2^{\mathbb{N}^*}$, where $\pi' = (v_1, v_2, \ldots, v_n) \in \mathbb{N}^*$ represents a 2-path $\pi = ((v_1, v_2), (v_2, v_3), \ldots, (v_{n-1}, v_n))$ of length $n-1$. For $a, b \in RPaths$ and $c \in \mathbb{B}$, we define:

- **Sum:** $a \oplus b = a \cup b$;
- **Product:** $a \otimes c = c \otimes a = \begin{cases} a & \text{if } c = 1 \\ \emptyset & \text{otherwise} \end{cases}$.

Let $A = (a_{ij})$, $B = (b_{ij})$ be matrices over $RPaths$ and $C = (c_{ij})$ be a Boolean matrix. We extend the operations to matrices:

- **Sum:** $A_{n\times m} \oplus B_{n\times m} = (a_{ij} \oplus b_{ij})_{n\times m}$;
- **Product:** $A_{n\times m} \otimes C_{m\times k} = (\bigoplus_{l=1}^{m} a_{il} \otimes c_{lj})_{n\times k}$;
- **Zero-matrix:** $0^{Paths}_{n\times k} = (\emptyset)_{n\times k}$.

3.5 Index Unary Operators

The path semiring alone is insufficient for enforcing path semantics (simple paths, trails). We introduce index unary operators that can filter paths based on vertex or edge repetition.

Definition 12 (Index unary operator). *Let $A_{n\times m} = (a_{ij})$ be a matrix over a semiring R. An index unary operator $I : R \times \mathbb{N} \times \mathbb{N} \mapsto R$ is applied element-wise: $I(A) = (I(a_{ij}, i, j))_{n\times m}$.*

For matrices over $RPaths$, we define specific index unary operators:

- $I_{Simple}(a, i, j) = \{(v_1, v_2, \ldots, v_n, j) \mid (v_1, v_2, \ldots, v_n) \in a, v_1 \neq v_n \text{ and } \forall 2 \leq t \leq n : v_t \neq j\}$;
- $I_{Trail}(a, i, j) = \{(v_1, v_2, \ldots, v_n, j) \mid (v_1, v_2, \ldots, v_n) \in a \text{ and } \forall 1 \leq t \leq n-1 : (v_t, v_{t+1}) \neq (v_n, j)\}$.

The I_{Simple} operator extends paths while filtering out those that would create vertex repetitions (enforcing simple paths). The I_{Trail} operator filters out extensions that would create edge repetitions (enforcing trails).

3.6 The Original LARPQ Algorithm for Reachability

We now present the original LARPQ algorithm for solving single-source 2-RPQ reachability, which serves as the foundation for our semiring generalization.

Input: $\mathcal{N} = \langle Q, \Sigma, \Delta, \lambda_{\mathcal{N}}, Q_S, Q_F \rangle$, $\mathcal{G} = \langle V, E, L, \lambda_{\mathcal{G}} \rangle$, $v_s \in V$
Output: Vector P^F of size $1 \times |V|$

1. $\{N^a\} \leftarrow$ Boolean decomposition of the $\mathcal{N}$ adjacency matrix
2. $\{G^a\} \leftarrow$ Boolean decomposition of the $\mathcal{G}$ adjacency matrix
3. $M_{|Q|\times|V|} \leftarrow \mathbf{0}_{|Q|\times|V|}$
4. $P_{|Q|\times|V|} \leftarrow \mathbf{0}_{|Q|\times|V|}$
5. For each $q \in Q_S$: set $M_{qv_s} \leftarrow 1$
6. While $M \neq \mathbf{0}$:
   - $M \leftarrow \bigoplus_{a \in \Sigma^{\leftrightarrow} \cap L^{\leftrightarrow}} ((N^a)^T \times M \times G^a)\langle \neg P \rangle$
   - $P \leftarrow P \oplus M$
7. $F_{1\times|Q|} \leftarrow \mathbf{0}_{1\times|Q|}$
8. For each $q \in Q_F$: set $F_{1q} \leftarrow 1$
9. Return $P^F = F \times P$

Fig. 1. Single-Source 2-RPQ Reachability using Linear Algebra

The algorithm maintains two matrices: M (the frontier) represents the relation between automaton states and graph vertices reached in the current step, while P accumulates all reachable pairs. The key insight is that on each iteration, $M_{qv} = 1$ if and only if there exists a 2-path in $\mathcal{G}$ from v_s to v and a path in $\mathcal{N}$ from some $q_s \in Q_S$ to q, both having the same labels and of length n (the iteration number). The mask $\langle \neg P \rangle$ ensures that each vertex-automaton state pair is processed only once, preventing infinite loops.

The correctness of the algorithm follows from the simultaneous breadth-first traversal of both the graph and the automaton. After termination, $P_{q_F v} = 1$ for $q_F \in Q_F$ if and only if there exists a 2-path from v_s to v whose label sequence belongs to the language accepted by $\mathcal{N}$.

The algorithm can also solve single-destination reachability by reversing the query. Let $\mathcal{R}^- = \{w^- = a_n^- a_{n-1}^- \dots a_1^- \mid w = a_1 a_2 \dots a_n \in \mathcal{R}\}$ and $\mathcal{N}^- = \langle Q, \Sigma, \Delta^-, \lambda_{\mathcal{N}}^-, Q_F, Q_S \rangle$. Then:

$$[[Q]]_{SDR} = [[\langle \mathcal{G}, \mathcal{R}^-, v_s \rangle]]_{SSR}.$$

Boolean matrix decomposition of $\mathcal{N}^-$ is obtained by transposing the matrices in the decomposition of $\mathcal{N}$.

4 A Semiring Framework for Single-Source RPQs

In this section, we present our main contribution: a generalized semiring-based framework for evaluating single-source 2-RPQs. The original LARPQ algorithm

operates over the Boolean semiring to solve reachability. We show that by replacing the Boolean semiring with other algebraic structures, the same algorithmic framework can solve different path-aggregation problems.

4.1 Generalized Algorithm

The key insight is that the LARPQ algorithm can be abstracted to work over any semiring equipped with appropriate operations. Algorithm 2 presents the generalized version.

Input: Semiring $\mathcal{S} = \langle S, \oplus, \otimes, \mathbf{0}, \mathbf{1} \rangle$, optional index unary operator I, 2-NFA $\mathcal{N} = \langle Q, \Sigma, \Delta, \lambda_{\mathcal{N}}, Q_S, Q_F \rangle$, graph $\mathcal{G} = \langle V, E, L, \lambda_{\mathcal{G}} \rangle$, source $v_s \in V$
Output: Result depending on semiring instantiation

1. $\{N^a\} \leftarrow \mathcal{S}$-matrix decomposition of $\mathcal{N}$
2. $\{G^a\} \leftarrow \mathcal{S}$-matrix decomposition of $\mathcal{G}$
3. $M \leftarrow \mathbf{0}$, $P \leftarrow \mathbf{0}$
4. For each $q \in Q_S$: $M_{qv_s} \leftarrow \mathbf{1}$
5. While $M \neq \mathbf{0}$:
   - If I defined: $M \leftarrow I(M)$
   - $P \leftarrow P \oplus M$
   - $M \leftarrow \bigoplus_{a \in \Sigma^{\leftrightarrow} \cap L^{\leftrightarrow}} ((N^a)^T \otimes M \otimes G^a)$
6. Return Export results from P based on Q_F

Fig. 2. Generalized Single-Source 2-RPQ using Semirings

The generalized algorithm differs from the original in several key aspects:

- Matrix operations ($\oplus$, $\otimes$) are performed over an arbitrary semiring $\mathcal{S}$;
- An optional index unary operator I can be applied to filter or transform path information;
- The final result format depends on the semiring instantiation.

We now present two important instantiations of this framework.

4.2 Instantiation I: Boolean Semiring for Reachability

By instantiating $\mathcal{S}$ with the Boolean semiring $\langle \{0, 1\}, \vee, \wedge, 0, 1 \rangle$ and omitting the index unary operator, we recover the original LARPQ algorithm for reachability (Algorithm 1). The matrix entries are Boolean values, where 1 indicates reachability. The mask operation $\langle \neg P \rangle$ ensures each vertex is processed only once.

The result is a Boolean vector P^F where $P^F_v = 1$ if and only if vertex v is reachable from the source via a path satisfying the regular language constraint.

4.3 Instantiation II: Path Semiring with Index Operators for All Paths

For enumerating all simple paths or all trails, we use the path semiring $\langle RPaths, \cup, \otimes, \emptyset, \{(s)\}\rangle$ combined with appropriate index unary operators.

When $I = I_{Simple}$, the algorithm enumerates all simple 2-paths satisfying the query. When $I = I_{Trail}$, it enumerates all trails. The key difference from reachability is that the frontier matrix M now stores actual path information (as sequences of vertices) rather than Boolean values. The index unary operator filters extensions that would violate the path semantics:

- I_{Simple} prevents extending a path with a vertex already present in the path (except the starting vertex), ensuring simplicity;
- I_{Trail} prevents extending a path with an edge already traversed, ensuring the result is a trail.

The algorithm maintains the BFS structure: on iteration n, M_{qv} contains all valid paths of length n from v_s to v that correspond to paths from some $q_s \in Q_S$ to q in the automaton.

Figure 3 illustrates an example of the algorithm step for the all-paths semantics.

[Figure: Placeholder]Algorithm step visualization for path semiring

Fig. 3. The algorithm step for graph and automaton for regular expression a^*b using path semiring.

4.4 Computational Complexity

We analyze the computational complexity of the generalized algorithm.

Complexity. Let $n = |V|$ be the number of vertices, $m = |E|$ be the number of edges, $k = |L|$ be the number of distinct labels, and $q = |Q|$ be the number of automaton states. For the Boolean semiring (reachability), the algorithm runs in $O(q \cdot nz(G) \cdot \text{iterations})$ time, where $nz(G)$ is the number of non-zero entries in the Boolean matrix decomposition and iterations $\leq n$. Using sparse matrix operations, the complexity is $O(q \cdot (\text{sparse matrix multiply}))$.

For the path semiring, the complexity is higher due to path storage:

- Space: $O(q \cdot n \cdot \text{max paths per pair})$;
- Time: Depends on the number of paths found, which can be exponential in the worst case (as the problem itself is #P-hard for general graphs).

The practical performance depends heavily on graph structure and query selectivity.

4.5 Proof of Correctness

We prove the correctness of the generalized semiring-based algorithm.

Frontier Invariant. After the t-th iteration of the main loop, for each automaton state $q \in Q$ and vertex $v \in V$:

$$M_{qv}^{(t)} = \{\pi \mid \pi \text{ is a 2-path from } v_s \text{ to } v \text{ of length } t \text{ satisfying the query}\}.$$

Accumulator Invariant. After the t-th iteration:

$$P_{qv}^{(t)} = \{\pi \mid \pi \text{ is a 2-path from } v_s \text{ to } v \text{ of length } \leq t \text{ satisfying the query}\}.$$

We now provide a formal proof of correctness using invariant-based reasoning.

Invariant 1 (Frontier). After iteration t ($t \geq 0$), for each automaton state $q \in Q$ and vertex $v \in V$:

$$M_{qv}^{(t)} = \{\pi \mid \pi \text{ is a 2-path from } v_s \text{ to } v \text{ of length exactly } t \text{ such that } \omega(\pi) \in L(\mathcal{N})\}.$$

Invariant 2 (Accumulator). After iteration t ($t \geq 0$), for each $q \in Q$ and $v \in V$:

$$P_{qv}^{(t)} = \{\pi \mid \pi \text{ is a 2-path from } v_s \text{ to } v \text{ of length } \leq t \text{ such that } \omega(\pi) \in L(\mathcal{N})\}.$$

When an index unary operator I is applied, the invariant is modified to consider only paths satisfying the semantic constraints enforced by I.

Proof of Invariant 1 (Frontier). We prove by induction on t.

Base case ($t = 0$): Initially, $M^{(0)}$ contains the empty path from v_s to v_s for each starting state $q \in Q_S$, represented by the identity element $\mathbf{1}$ of the semiring. Formally:

$$M_{qv}^{(0)} = \begin{cases} \mathbf{1} & \text{if } q \in Q_S \text{ and } v = v_s \\ \mathbf{0} & \text{otherwise} \end{cases}$$

This matches the definition in Invariant 1 for paths of length 0.

Inductive step: Assume Invariant 1 holds after iteration t. Consider iteration $t + 1$. The update step computes:

$$M^{(t+1)} = \bigoplus_{a \in \Sigma^{\leftrightarrow} \cap L^{\leftrightarrow}} (N^a)^T \otimes M^{(t)} \otimes G^a.$$

For a fixed label a, the product $(N^a)^T \otimes M^{(t)}$ extends each path in $M^{(t)}$ by one automaton transition labeled a. Specifically, if $M_{q'v'}^{(t)}$ contains a path π, then $((N^a)^T \otimes M^{(t)})_{qv'}$ contains the extended path for each predecessor state q with a transition $q \xrightarrow{a} q'$ in $\mathcal{N}$.

Multiplying by G^a then extends these paths by one graph edge labeled a. The result contains all paths of length $t+1$ from v_s to each vertex v that correspond to paths in the automaton from some starting state to q.

Taking the union (via $\bigoplus$) over all labels a combines all possible extensions, yielding exactly the set of paths of length $t+1$ satisfying the query. This establishes Invariant 1 for iteration $t+1$.

When an index unary operator I is applied, it filters or transforms the paths in $M^{(t+1)}$ according to the semantic constraints. The correctness of filtering follows directly from the definitions of I_{Simple} and I_{Trail}.

Proof of Invariant 2 (Accumulator). Invariant 2 follows directly from Invariant 1 and the accumulation step:

$$P^{(t+1)} = P^{(t)} \oplus M^{(t+1)}.$$

By the induction hypothesis, $P^{(t)}$ contains all paths of length $\leq t$. Adding $M^{(t+1)}$ (which contains paths of exactly length $t+1$ by Invariant 1) yields all paths of length $\leq t+1$.

Theorem (Correctness). Let $\mathcal{Q} = \langle \mathcal{G}, \mathcal{R}, v_s \rangle$ be a 2-RPQ and let the algorithm terminate after T iterations. Then:

1. For Boolean semiring: $P_v^F = 1$ iff $v \in [[Q]]_{SSR}$;
2. For path semiring with I_{Simple}: the result contains exactly all 2-simple paths in $[[Q]]_{SSASP}$;
3. For path semiring with I_{Trail}: the result contains exactly all 2-trails in $[[Q]]_{SSAT}$.

Proof. By Invariant 2, after termination (when $M = \mathbf{0}$), P contains all paths from v_s to each vertex that satisfy the query. The export step multiplies by F, which selects only paths ending in final automaton states.

For Boolean semiring, $P_{q_F v} = 1$ for some $q_F \in Q_F$ iff there exists a path from v_s to v whose label is accepted by the automaton. This is precisely the definition of reachability.

For the path semiring, I_{Simple} ensures that only simple paths remain (no vertex repetition), and I_{Trail} ensures only trails remain (no edge repetition). The index operators are applied at each step, filtering out non-compliant extensions immediately, guaranteeing correctness.

[Figure: Placeholder]Detailed algorithm step showing path filtering

Fig. 4. Frontier update with path semantics: (a) simple paths filter, (b) trails filter.

5 Evaluation

In this section, we present the experimental evaluation of our implementation. Few words about implementation details. We compare the performance of the proposed semiring-based LARPQ algorithm against state-of-the-art graph database management systems and other linear algebra-based approaches.

5.1 Experimental Setup

Hardware

We evaluate the reachability semantics (Boolean semiring instantiation) on real-world knowledge graphs. The experiments are designed to answer the following research questions:

- **RQ1:** How does our implementation compare with existing linear algebra-based RPQ solutions?
- **RQ2:** How does our implementation compare with production graph database systems?
- **RQ3:** What is the performance behavior across different query types and graph structures?

Competitors We compare against the following systems:

- **MillenniumDB** [**?**]: A persistent, open-source graph database with state-of-the-art RPQ performance;
- **FalkorDB** [**?**]: An in-memory property-graph database using GraphBLAS;
- **RPQ-matrix** [**?**]: Another linear algebra-based RPQ evaluation algorithm.

Datasets We use the following real-world knowledge graphs:

- **Wikidata**: [Detailed statistics to be added]
- **YAGO**: [Detailed statistics to be added]

Queries [Query set description to be added]

Table 1. Dataset Statistics

Dataset	$\|V\|$	$\|E\|$	$\|L\|$
Wikidata	!!!	!!!	!!!
YAGO !!!			

5.2 Experimental Results

Experimental results will be presented here. Based on preliminary findings from our earlier work [**?**], we expect the following:

- The proposed algorithm should demonstrate significant speedups over existing linear algebra-based solutions (RPQ-matrix) on complex queries involving Kleene star operators;
- The algorithm should achieve comparable or better performance than MillenniumDB on reachability queries;

– Performance should be competitive with FalkorDB while supporting a broader class of regular path queries.

Detailed experimental results including:

– Tables comparing execution times, memory consumption, and timeout rates;
– Scatter plots showing per-query performance distributions;
– Analysis of performance across different query types (simple patterns vs. Kleene star, Cypher-compatible vs. non-compatible);
– Scalability experiments on larger graphs.

will be included in the final version of this paper.

Table 2. Expected Performance Summary (Preliminary)

Metric	**LA 2-RPQ**	**MillenniumDB**	**RPQ-Matrix**	**FalkorDB**
Mean time (ms)	µ	µ	µ	µ
Median time (ms)	µ	µ	µ	µ
Speedup vs LA	1.0	µ	µ	µ
Memory (GB)	µ	µ	µ	µ
Timeouts	µ	µ	µ	µ

µ= to be determined based on experiments

[Figure: Placeholder]Query performance scatter plot

Fig. 5. Point-wise query execution times comparison.

The experimental evaluation will demonstrate that our semiring-based approach provides a flexible framework for solving various path problems while maintaining competitive performance with specialized solutions.

6 Related Work

Regular path queries have been extensively studied in the context of graph databases. In this section, we discuss related work on linear algebra-based graph analytics, RPQ evaluation techniques, and semiring-based path problems.

Linear Algebra-Based Graph Analytics. The GraphBLAS initiative [**?**,**?**] established a standardized interface for expressing graph algorithms using linear algebra operations. LAGraph [**?**] provides a collection of graph algorithms built on GraphBLAS, including BFS, PageRank, and triangle counting. Our work extends this ecosystem with RPQ algorithms.

The use of sparse matrices for graph representation has proven effective for large-scale graphs. Real-world graphs exhibit sparsity, allowing for memory-efficient storage and fast computation using formats like CSR/CSC [**?**]. The GraphBLAS specification supports arbitrary semirings, enabling the expression of algorithms that go beyond traditional numeric computations.

Recent work has explored linear algebra approaches for various graph problems. However, RPQ evaluation in this paradigm remains underexplored, with only a few prior works [**?**,**?**] addressing this problem.

RPQ Evaluation Methods. Traditional approaches to RPQ evaluation include:

- **Automaton-based methods:** Construct the product automaton of the graph and the query NFA, then perform traversal [**?**]. These methods can be combined with indexing techniques for improved performance.
- **BFS-based methods:** Perform breadth-first search guided by the query automaton [**?**]. The original LARPQ algorithm [**?**] falls into this category.
- **Matrix-based methods:** Translate the regular expression into matrix operations. The RPQ-matrix algorithm [**?**] uses this approach, representing adjacency matrices in compressed formats like k2-trees or CSR/CSC.

Our approach combines elements from both BFS and matrix-based methods, using the simultaneous traversal of graph and automaton structures while leveraging sparse matrix operations for efficiency.

Different query semantics have been studied in the literature. The all-shortest-paths semantics is addressed in [**?**], while all-simple-paths and all-trails semantics are more challenging due to the potential for exponential result sizes. Our framework provides a unified approach for handling these different semantics.

Semiring-Based Path Problems. The algebraic approach to path problems has a long history. Kleene's algorithm [**?**] computes reachability using matrix operations over the Boolean semiring. This approach has been generalized to arbitrary semirings for solving various algebraic path problems [**?**,**?**].

The connection between semirings and path enumeration is well-established. The tropical semiring (min-plus) corresponds to shortest path problems, while the Boolean semiring corresponds to reachability. Our work extends this connection to include the path semiring for explicit path enumeration.

Provenance semirings [**?**] provide a framework for tracking the origin of query results. Our path semiring can be seen as a form of provenance tracking, where each result path is explicitly stored.

Context-free path querying (CFPQ) has also been approached using linear algebra [**?**]. While CFPQ is more expressive than RPQ, our work focuses on regular languages and different evaluation semantics.

Graph Database Systems. Modern graph database systems vary significantly in their RPQ support:

- **MillenniumDB** [**?**]: A persistent graph database with state-of-the-art RPQ performance, supporting various path semantics.
- **FalkorDB** [**?**] (formerly RedisGraph): An in-memory property-graph database that uses GraphBLAS for query evaluation. Supports a subset of regular path queries through Cypher.
- **Neo4j:** The most widely used graph database, supporting Cypher queries including limited forms of RPQ.
- **Blazegraph:** An RDF database with SPARQL support including property paths.

Our implementation differs from these systems in its focus on the linear algebra paradigm and its flexibility in supporting different semirings for various path semantics.

Table 3 summarizes the comparison of our approach with related systems.

[Table: Placeholder]Comparison of RPQ approaches

Table 3. Comparison of RPQ evaluation approaches. LA = Linear Algebra, AB = Automaton-based, BFS = Breadth-first search.

Our contributions differ from prior work in several key aspects:

- We provide a **unified framework** that can solve multiple path semantics using different semiring instantiations;
- We extend the original LARPQ algorithm beyond reachability to path enumeration;
- We leverage the **GraphBLAS standard** for high-performance sparse matrix operations.

7 Conclusion and Future Work

We have presented a semiring-based generalization of the linear algebra approach for evaluating two-way regular path queries. The experimental evaluation demonstrates that the proposed approach achieves competitive performance with state-of-the-art graph database systems while providing a more flexible framework for different path semantics.

Future work includes the use of additional semirings, such as the tropical semiring for shortest-path queries and counting semirings for path counting, as well as integration with property graphs and more expressive query languages (e.g., CRPQs). Another direction is a distributed implementation (e.g., using CombBLAS [**?**]) and acceleration by offloading matrix operations to GPUs (e.g., using GraphBLAST, cuBool, Spla [**?**]).

8 Availability Statement

Source code and additional materials are available at https://github.com/SparseLinearAlgebra/larpq. Evaluation framework.

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